Lecture 5: Time Series Approaches to Identifying Macroeconomic Shocks

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Motivation

- We have seen: structural business-cycle models admit **SVMA representations**

$$y_t = \sum_{\ell=0}^{\infty} \Theta_\ell \varepsilon_{t-\ell}$$

- Next few classes: how can we use data to learn about the Θ's? Three approaches:

  1. **Semi-structural, time series**: estimate second moments of y_t + use economic identifying assumptions that hold for *families* of models. This will be our focus for the next few lectures.
  2. **Semi-structural, cross-sectional**: what can microeconometric causal designs teach us? Will discuss this towards the end of the course.
  3. **Structural**: specify full-blown model, then *estimate* this model using micro/macro data. Will discuss throughout. Needed when empirical/semi-structural approaches fall short.

Outline

1. The Time Series Identification Challenge

2. Invertibility + X
   - Exact Exclusion Restrictions
   - Sign Restrictions
   - Other Approaches
   - Degree of Invertibility

3. Identification without Invertibility
   - Instruments/Proxies
   - Recoverability

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Framework & identification challenge

- **Assumption**: SVMA(∞) model for macroeconomic observables

$$y_t = \sum_{\ell=0}^{\infty} \Theta_\ell \varepsilon_{t-\ell}, \quad \varepsilon_t \sim WN(0, I) \tag{1}$$

So y_t is covariance-stationary. Note that we may well have more shocks than observables, i.e., $n_\varepsilon > n_y$.

- Our **Q**: what can we say about ε_t and the Θ_ℓ's from time-series data on y_t?
  - Treat **second-moment properties** of y_t—i.e., autocovariance function $\Gamma_y(\bullet)$—as known
    - → Recall notation: $\Gamma_y(\ell) = \text{Cov}(y_t, y_{t-\ell})$, i.e., $n_y \times n_y$ matrix for $\ell = 0, 1, 2, \ldots$
    - → For covariance-stationary y_t we can consistently estimate $\Gamma_y(\bullet)$, w/o any structural asn's

    Aside: Will not exploit higher-order moments. Will briefly discuss the lit. that does that.
  - **Problem**: as we will see, infinitely many—and economically very different—SVMAs are consistent with $\Gamma_y(\bullet)$

Wold decomposition

- Let's first of all show that we can always find *a* **possible VMA representation**
- We can do so easily using the **Wold decomposition** of y_t:

$$y_t = \tilde{\Psi}(L)\tilde{\varepsilon}_t, \tag{2}$$

where $\tilde{\varepsilon}_t \equiv \Sigma^{-1/2} u_t \sim WN(0, I)$ and

$$u_t \equiv y_t - \mathbb{E}^*(y_t \mid \{y_\tau\}_{-\infty<\tau<t}), \quad u_t \sim WN(0, \Sigma).$$

  - Recall notation: $\mathbb{E}^*(\bullet \mid \bullet)$ denotes *linear projection* on the indicated information set
  - Interpretation: there is nothing "structural" about the $\tilde{\varepsilon}_t$'s—they are just **orthogonal innovations** in a representation of y_t, scaled to have unit variance

- Next let's show that there exist **many** such observationally equivalent representations, none of which necessarily have any **economic meaning** ...

Indeterminacy I: rotations

- The first kind of indeterminacy is what I'll refer to as **static indeterminacy**
  - Start with the true SVMA model, and assume that $n_\varepsilon = n_y = n$. Now let $Q \in O(n)$ denote an $n \times n$ orthogonal matrix. Then the process

$$y_t = \sum_{\ell=0}^{\infty} \tilde{\Theta}_\ell \tilde{\varepsilon}_{t-\ell}, \quad \tilde{\varepsilon}_t \sim WN(0, I),$$

  with $\tilde{\Theta}_\ell = \Theta_\ell Q'$ and $\tilde{\varepsilon}_t = Q\varepsilon_t$ has the exact **same second-moment properties**:

$$\text{Cov}(y_t, y_{t-h}) = \sum_{\ell=0}^{\infty} \Theta_{\ell+h}\Theta_\ell' = \sum_{\ell=0}^{\infty} \tilde{\Theta}_{\ell+h}\tilde{\Theta}_\ell'.$$

- In words: **orthogonal rotations** of today's shocks leave covariances unchanged

  Problem: can just relabel the shocks, or take linear combinations—data won't be able to reject it.

Indeterminacy II: root flipping

- The second kind of indeterminacy is what I'll refer to as **dynamic indeterminacy**
  - To illustrate this consider a simple univariate MA(1), i.e. $n_y = n_\varepsilon = 1$:

$$y_t = \theta_0 \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

  - Now compare this with the following alternative process:

$$y_t = \theta_1 \varepsilon_t + \theta_0 \varepsilon_{t-1}$$

    Note how they share the same variances and covariances! But of course the implied shock IRFs look very different as long as $\theta_0 \neq \theta_1$ ...

  - What did we do? We took the polynomial $\Theta(L) = \theta_0 + \theta_1 L$ and **"flipped the root"**

- General lesson: **flipping roots** of MA polynomials leaves second moments unchanged

  For a general multivariate treatment of this see Lippi and Reichlin (1994), Plagborg-Møller (2019).

Indeterminacy III: system size

- The third kind of indeterminacy is what I'll refer to as **size indeterminacy**
  - So far $n_y = n_\varepsilon$, but options for SVMA processes expand even more if we allow $n_\varepsilon > n_y$
  - To illustrate suppose again that the true model is the same one-shock MA(1) as before, but now we entertain the alternative ***two*-shock process**

$$y_t = \theta_{01}\varepsilon_{1t} + \theta_{02}\varepsilon_{2t} + \theta_{11}\varepsilon_{1,t-1} + \theta_{12}\varepsilon_{2,t-1}$$

  - This process is just as consistent with the autocovariance f'n of the data as long as

$$\begin{aligned} \text{Var}(y_t) &= \theta_{01}^2 + \theta_{02}^2 + \theta_{11}^2 + \theta_{12}^2 = \theta_0^2 + \theta_1^2 \\ \text{Cov}(y_t, y_{t-1}) &= \theta_{01}\theta_{11} + \theta_{02}\theta_{12} = \theta_0\theta_1 \end{aligned}$$

    But of course it implies very different shocks and impulse response functions ...

- Lesson: we can always split any given n_y-dimensional autocovariance structure into many more $(n_\varepsilon > n_y)$ **distinct shocks**

Summary: identification challenge

- **Takeaway**: from second moments alone, the SVMA model is severely under-identified
  Of course that's unsurprising. We haven't made any economic identifying assumptions yet!
- Now: what kind of addt'l **identifying assumptions** does the time series lit. impose?
  Will go from Sims' (1980, ECMA) classic, through the VAR literature, to the recent LP literature.

1. **Invertibility + addt'l exclusion restrictions**
   This is my discussion of what the classical "VAR" literature is doing, though as we'll see it's not really about "VARs" per se.
2. **Identification w/o invertibility – macro IVs**
   The "macro IV" approach is very popular these days, and looks rather similar to by-now standard microeconometric/"credibility revolution" practice.

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Invertibility

- Recall our definition of **invertibility**:

Definition

A VMA process $\{y_t\}$ is said to be **invertible** with respect to $\{\varepsilon_t\}$ if

$$\varepsilon_t \in \text{span}(y_\tau, -\infty < \tau \leq t)$$

- This is an economically **substantive assumption**:
  - It implies that an econometrician observing $\{y_\tau\}_{-\infty<\tau\leq t}$ can predict just as well as an agent living in the model and observing $\{\varepsilon_\tau\}_{-\infty<\tau\leq t}$
  - Certainly need $n_\varepsilon \leq n_y$. The more (relevant) data we include, the more plausible this as'n.
  - It can be violated for standard macro models with standard observables, even if $n_\varepsilon = n_y$. Let's consider one very simple example. Following Fernández-Villaverde et al. (2007).

Invertibility: illustration

- Consider a simple **permanent-income consumption model**:
  - Income e_t and consumption c_t are given as
$$e_t = \sigma_\varepsilon \varepsilon_t, \quad c_t = c_{t-1} + \frac{\bar{r}}{1+\bar{r}} \sigma_\varepsilon \varepsilon_t$$
  - Also suppose we observe savings growth $y_t \equiv \Delta s_t$, where $s_t = e_t - c_t$:
$$y_t = \frac{1}{1+\bar{r}} \sigma_\varepsilon \varepsilon_t - \sigma_\varepsilon \varepsilon_{t-1}$$
- Is this model **invertible**?
  - We have one shock and one observable, so things are looking good
  - But unfortunately it doesn't work out, since the backward sum diverges:
$$\varepsilon_t = \frac{1}{\sigma_\varepsilon} \sum_{\ell=0}^{T} (1+\bar{r})^{\ell+1} y_{t-\ell} + (1+\bar{r})^{T+1} \varepsilon_{t-T-1}$$

Note: could instead back out the current shock by solving *forward*.

Invertibility: discussion

- More general point: **invertibility** can fail for many reasons
  - A necessary condition is that we have (at least) as many observables as shocks
  - But even that is not sufficient. Key culprits for non-invertibility when $n_\varepsilon = n_y$ are:
    a) There are news shocks (e.g., forward guidance, fiscal spending plans)
    Like our MA from the previous slide—larger lagged than contemporaneous effect naturally generates non-invertibility. See Leeper et al. (2013) for a detailed discussion.

    Extreme example: far-ahead TFP news shock that doesn't move observables today.

    b) There are noise shocks (e.g., noise that is misperceived as news about future fundamentals)
    See Chahrour-Jurado (2018) for a detailed discussion.

  Would need to look into the *future*—when the news realizes, and to know whether it was news or noise. $n_y = n_\varepsilon$ usually only ensures $\varepsilon_t \in \text{span}(y_\tau, -\infty < \tau < \infty)$, not that the *past* is enough.

- So what do we gain from this **strong assumption**? As it turns out, a lot ...

Invertibility & Wold innovations

- **Invertibility** has important implications for **Wold innovations**
  - Recall that
$$u_t \equiv y_t - \mathbb{E}^*(y_t \mid \{y_\tau\}_{-\infty<\tau<t})$$
  - Since y and u share the same span, it follows that, *under invertibility*,
$$\varepsilon_t \in \text{span}(u_\tau, -\infty < \tau \leq t)$$
    But ε_t is uncorrelated with all u_τ for $\tau < t$, and u_t is white noise, so $\varepsilon_t \in \text{span}(u_t)$ or
$$H\varepsilon_t = u_t$$
    for some matrix H, where $HH' = \Sigma = \text{Var}(u_t)$.
- But the **Wold innovations** u_t and their "causal effects" $\Psi(L)$ are easily estimable. We have thus made a lot of progress ...

Invertibility & Wold innovations

- By imposing **invertibility**, we have been able to identify the structural shocks ε_t and their causal effects up to the **static rotation problem**:

  1. The structural shocks are given as

  $$\varepsilon_t = Q\Sigma^{-1/2}u_t = Q\tilde{\varepsilon}_t$$

  2. Their dynamic causal effects are given as

  $$\Theta(L) = \Psi(L)\Sigma^{1/2}Q'$$

- **Invertibility** has thus helped by eliminating the **dynamic & size indeterminacy**
  - Intuition: $n_\varepsilon \leq n_y$ (which is required for invertibility) rules out size indeterminacy, and then the sufficiency of *past* observables rules out the dynamic MA root flipping
  - Remains to find identifying assumptions to pin down Q [= static indeterminacy] ...

Connection to VARs

- Note that this **approach to identification** can be *operationalized* using **VARs**:
  - We have the SVMA $y_t = \Theta(L)\varepsilon_t$. Under invertibility, the inverse $\tilde{A}(L) \equiv \Theta(L)^{-1}$ exists.
  - Letting $A(L) = H\tilde{A}(L)$, we thus have

$$A(L)y_t = u_t = H\varepsilon_t.$$

    That is, **VAR residuals** = **Wold residuals**, which we now just need to rotate.

- The derivations above are why the literature often speaks of **"VAR identification"**
  - But that's a bit misleading. ID asn's are **invertibility** + **something else** to pin down Q.
  - A VAR is just one possible—but not the only—**estimation strategy**. Will discuss later.

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Exact exclusion restrictions

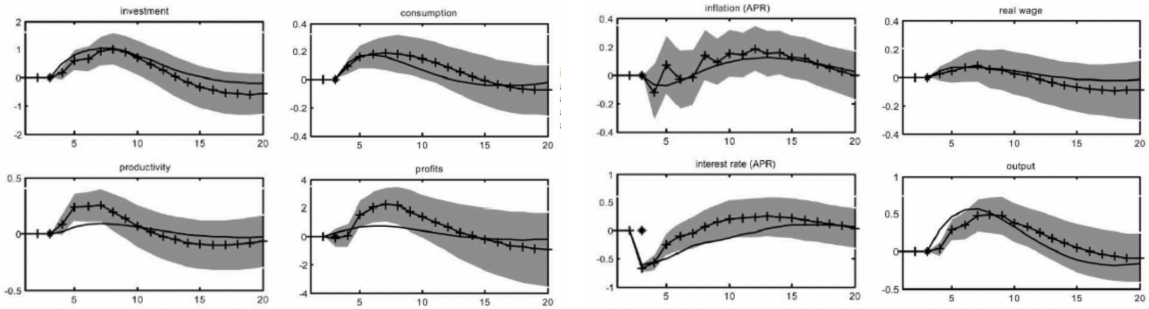
So how can we identify the **right rotation matrix** Q?

- Traditional approach: point-identify the system through *exact* zero restrictions on $\Theta(L)$
  - Need to impose enough restrictions such that only one $Q \in O(n)$ is consistent with them
- We will consider some of the most well-known examples of such restrictions:
  1. Restrictions on **short-run impulse responses** Θ_0
     We will review Christiano-Eichenbaum-Evans (1999, 2005).
  2. Restrictions on **long-run impulse responses** $\sum_{\ell=0}^{\infty} \Theta_\ell$
     We will review Blanchard-Quah (1989).

Short-run identification

- Consider the assumption that Θ_0 is **lower-triangular**:
  - In economic terms: variable 1 only responds contemporaneously to shock 1, variable 2 responds contemporaneously to shocks 1 & 2, and so on
  - Mathematically: $H = \text{chol}(\Sigma)$
- Christiano et al. (1999, 2005) operationalize this as'n to identify **monetary shocks**
  - Observables y_t: real GDP, real consumption, GDP deflator, real investment, real wage, labor productivity, federal funds rate, real profits, growth rate of M2
  - Assumptions: invertibility + recursive shock ordering
    Here this means that all variables listed before the fed funds rate don't respond within the period to monetary shocks, while monetary policy doesn't respond to changes in the variables listed after.
- This identification approach gave the "canonical" monetary policy IRFs *[next slide]*
  Our interpretation (Lecture Note 4): rate path is "treatment," rest is causal effects.

Short-run identification



Can you think of any problems with this?

Long-run identification

- Suppose we are willing to impose that the **long-run response** of variable i to shock j is zero. If the variable is in growth rates, then this implies a zero *cumulative* IRF:

$$0 = \sum_{\ell=0}^{\infty} \Theta_{i,j,\ell} = \sum_{\ell=0}^{\infty} \Psi_{i,\bullet,\ell} H_{\bullet,j} = \sum_{\ell=0}^{\infty} \Psi_{i,\bullet,\ell} \Sigma^{1/2} Q'_{j,\bullet} \tag{3}$$

  - (3) thus imposes an additional restriction on the orthogonal matrix Q, aiding with identification of the system

- Blanchard-Quah (1989) use this as'n to disentangle **demand & supply shocks**
  - Observables y_t: output growth, unemployment
  - Assumptions: invertibility + demand shocks do not affect real output in the long run

Long-run identification

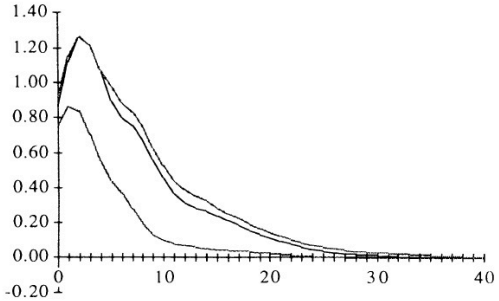


FIGURE 3. OUTPUT RESPONSE TO DEMAND

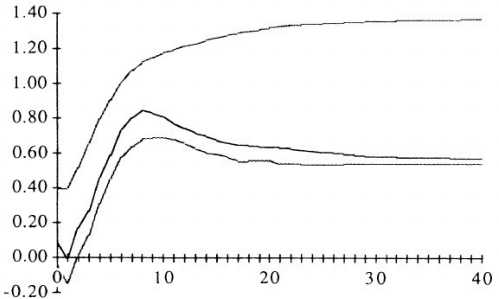


FIGURE 4. OUTPUT RESPONSE TO SUPPLY

Can you think of any problems with this?

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Motivation

- **Exact** zero restrictions are widely regarded as implausible

  Effects of mon. policy on output may be delayed, but not exactly 0 on impact. This is not innocuous: What if other shocks also have *near*-zero impact effects? Are monetary shocks closest to zero?

- One popular alternative: impose restrictions on **signs** of impulse responses [Uhlig (2005)]
  - Example: a contractionary monetary shock should probably lead to an increase in interest rates and a decrease in prices (eventually)
  - This is promising: if central banks lean against inflation, then all non-policy shocks lead to a *positive* contemporaneous co-movement of interest rates and inflation

- **Q**: How can we operationalize this intuition? What can invertibility + sign restrictions tell us about the **SVMA model**?

Overview

- **Implementation overview**
  - Recall that IRFs for a given rotation Q are equal to

$$\Theta(L) = \Psi(L)H = \Psi(L)\Sigma^{1/2}Q'$$

  - Suppose we want to impose that $\Theta_{i,j,\ell} \geq 0$. This **rules out** some Q's! We need

$$\Psi_{i,\bullet,\ell}\Sigma^{1/2}Q'_{j,\bullet} \geq 0 \tag{4}$$

$\Longrightarrow$ identified set: keep all Q's such that (i) $QQ' = I$ and (ii) all imposed sign restrictions hold

- **Discussion**
  - Sign restrictions alone are *weak* identifying information. Significant results in earlier work often driven by unacknowledged addt'l restrictions on Q. Detailed discussion on next slides.
  - In practice nowadays: often combine with some zero (or close-to-zero) restrictions

Example: sign restrictions for monetary policy

- Let's illustrate the workings of sign restrictions through our familiar **toy model**:

$$
\begin{aligned}
y_t &= \mathbb{E}_t\left(y_{t+1}\right) - \frac{1}{\gamma}\left(i_t - \mathbb{E}_t\left(\pi_{t+1}\right)\right) + \sigma^d \varepsilon_t^d && \text{(IS)} \\
\pi_t &= \kappa y_t + \beta \mathbb{E}_t\left(\pi_{t+1}\right) - \sigma^s \varepsilon_t^s && \text{(NKPC)} \\
i_t &= \phi_\pi \pi_t + \sigma^m \varepsilon_t^m && \text{(TR)}
\end{aligned}
$$

As usual the shocks satisfy $\varepsilon_t \equiv (\varepsilon_t^d, \varepsilon_t^s, \varepsilon_t^m) \sim WN(0, I)$.

- As we've seen, solving the model gives a **static mapping** from shocks to observables:

$$
\begin{pmatrix} y_t \\ \pi_t \\ i_t \end{pmatrix} = \underbrace{\begin{pmatrix} + & + & - \\ + & - & - \\ + & - & + \end{pmatrix}}_{\Theta} \times \begin{pmatrix} \varepsilon_t^d \\ \varepsilon_t^s \\ \varepsilon_t^m \end{pmatrix}
$$

Denote the response of variable j to shock k by Θ_{jk}.

Example: sign restrictions for monetary policy

- Interpreting the Q's
  - Let $x_t = (y_t, \pi_t, i_t)'$ and $\Sigma_x = \text{Var}(x_t)$. Note that

$$\Sigma_x \equiv \text{Var}(x_t) = \Theta\Theta'$$

  - Before we defined the Q's relative to $\Sigma_x^{1/2}$. More generally we can define them w.r.t. any matrix A such that $AA' = \Sigma_x$, including Θ. The Q's are just rotations w.r.t. a basis.
  - Choosing Θ as the basis has an advantage: if $\tilde{\Theta} = \Theta Q'$, then the matrix Q links the true shocks ε_t to the mis-identified shocks $\tilde{\varepsilon}_t$

$$\tilde{\varepsilon}_t \equiv Q\varepsilon_t$$

- We'd like to find identifying assumptions that are consistent with $Q = I$ (so $\tilde{\varepsilon}_t = \varepsilon_t$) but rule out (almost) everything else. Will **sign restrictions** on Θ do the trick?

Example: sign restrictions for monetary policy

- Let's impose the following **minimalist sign restrictions**:

$$\Theta = \begin{pmatrix} ? & ? & ? \\ ? & ? & - \\ ? & ? & + \end{pmatrix} \tag{5}$$

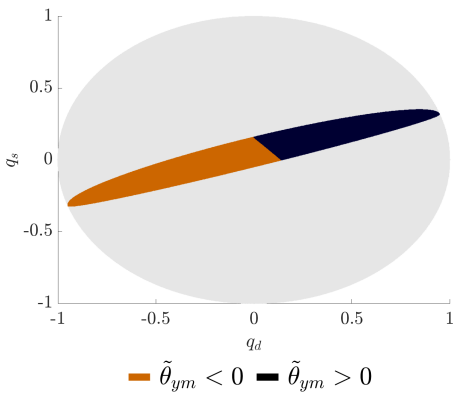
- Is this enough to learn anything about the **real effects of monetary policy**, Θ_{ym}?
  - Promising: monetary shocks are the *only* shocks to move π_t and i_t in opposite directions
  - This implies that $\tilde{\varepsilon}_t^m = \varepsilon_t^m$ is consistent with (5), while $\tilde{\varepsilon}_t^m = \varepsilon_t^d$ or $\tilde{\varepsilon}_t^m = \varepsilon_t^s$ are not
- Formally: (5) gives us an *identified set* $[\underline{\Theta}_{ym}, \overline{\Theta}_{ym}]$ for Θ_{ym}, defined via the programs

$$\inf_q / \sup_q \quad \Theta_{1,\bullet} \cdot q$$

s.t. $||q|| = 1$ and $\Theta_{2,\bullet} \cdot q \leq 0$, $\Theta_{3,\bullet} \cdot q \geq 0$

The identified set & masquerading shocks

What does the **identified set** $[\underline{\Theta}_{ym}, \overline{\Theta}_{ym}]$ look like?



- Since Θ is invertible, it's easy to see that we can't sign Θ_{ym}:

$$\underline{\Theta}_{ym} < 0 < \overline{\Theta}_{ym}$$

- Interpretation: "masquerading" shocks
  - The truth ($q = (0, 0, 1)'$) is in the identified set, while $q = (1, 0, 0)'$ and $q = (0, 1, 0)'$ are not
  - But: linear combos of expansionary supply and demand shocks can also move $i \uparrow$ & $\pi \downarrow$, but $y \uparrow$
- Typically, the identified set for Q will contain multiple elements, as seen here.

Econometric aside: the Haar prior

- Much of the (early) sign restrictions literature actually doesn't report **identified sets**
- Instead: additionally impose a **prior on orthogonal rotation matrices**, and then characterize the implied posterior
  - Common choice (for computational reasons): "uniform" Haar prior
  - The effect of this prior is to re-weight the parameters in the identified set. Desirable?
- Briefly: no. The prior is **funky in economic terms** and can be central to posterior tightness in actual applications. See Baumeister-Hamilton (2015) and Wolf (2020).
  - Best practice: frequentist inference on entire identified set or prior-robust inference
    Moon-Schorfheide (2012), Giacomini-Kitagawa (2021)
  - Alternatively: explicitly discuss & defend additional probabilistic prior over $O(n)$
    As done in Baumeister-Hamilton (2015) or Plagborg-Møller (2019).

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Literature overview

What other approaches for **identification under invertibility** are there?

1. Combining **zero and sign restrictions** Arias, Rubio-Ramírez and Waggoner (2018)
   - Can yield much tighter identified sets than sign restrictions alone, even without additional imposition of the Haar prior
2. Sign/magnitude restrictions on **other objects**, not IRFs. Examples:
   - Shock that explains a large share of fluctuations in variable X Angeletos et al. (2020)
   - Shock that explains most volatility at a particular date Antolín-Díaz & Rubio-Ramírez (2018)
3. Explicitly defended **probabilistic priors on** Q (rather than the Haar prior) Baumeister and Hamilton (2015), Plagborg-Møller (2019)
4. Use more information than just second moments (= **"statistical" identification**)

Statistical identification

- Identification through **higher-order moments** has recently gained in popularity
- Basic idea:
  - Lack of identification so far because we only looked at second moments. If shocks are not normal, higher-order moments can help
    1. Next slide: identification through heteroskedasticity point-identifies the model
       To me the most promising of these "statistical" approaches, since there's still a lot of economics in trying to come up with reasons for why relative shock volatilities change.
    2. Alternative: if true shocks are independent and non-normal, then also point-identified
       Intuition: only the true rotation of the u_t's is independent. See Gouriéroux et al. (2017).
  - Note: even with statistical id, you still need economic reasoning to *label* the shocks
    E.g.: which of the "statistically" identified shocks is the monetary policy shock?

Identification through heteroskedasticity in one slide

- Suppose now that the struct. shocks ε_t have different volatilities on two subsamples
  - Write the variances as $\{(\sigma_j^a)^2, (\sigma_j^b)^2\}_{j=1}^n$ for our two subsamples a and b [Note: now we are not normalizing the shock variances, so we instead normalize the diagonal entries of H to 1.]
  - For each sub-sample $k = a, b$, the Wold residuals u_t^k satisfy

$$\text{Var}(u_t^k) = H \begin{pmatrix} (\sigma_1^k)^2 & \dots & 0 \\ & \ddots & \\ 0 & \dots & (\sigma_n^k)^2 \end{pmatrix} H' = \Sigma_k$$

- We thus now have more information that we can exploit:

$$\Sigma_b \Sigma_a^{-1} = H \begin{pmatrix} (\frac{\sigma_1^b}{\sigma_1^a})^2 & \dots & 0 \\ & \ddots & \\ 0 & \dots & (\frac{\sigma_n^b}{\sigma_n^a})^2 \end{pmatrix} H^{-1}$$

  - Columns of H are eigenvectors of $\Sigma_b \Sigma_a^{-1}$, which is identified. Given our normalization of the diagonal of H to 1, we see that H is identified if the volatility ratios differ across shocks

Graphical intuition

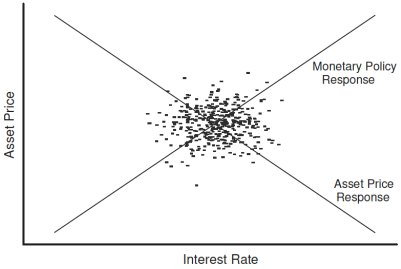


Fig. 1. Joint determination of interest rates and asset prices.

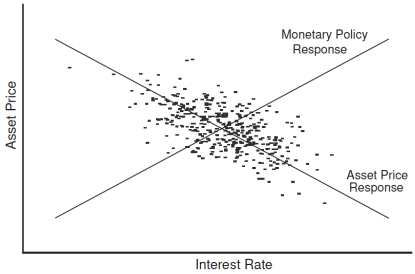


Fig. 2. Policy dates.

- Key idea: only two slopes can jointly rationalize these two clouds
  - Easiest to see if one shock is *dominant* in one regime = event study. But not necessary.
  - Of course then still need to label the two lines = two shocks
    Here: what is causal effect of MP and what is rule response? Other natural application: what's demand and what's supply?

Some thoughts on statistical identification

- Comparison with our **standard identification arguments**
  - Classical invertibility-based analysis assumes white noise shocks, e.g. allowing for stochastic volatility. Just doesn't exploit higher-order moments
  - Statistical approaches achieve identification by strengthening assumptions on the shock process: independence/conditional orthogonality rather than just white noise
- Some **words of caution**: See Montiel-Olea et al. (2022) + Kolesár and Plagborg-Møller (2025).
  - Methods *heavily* leverage that system is linear. "Anything goes"-type results if the system is allowed to be non-linear. Also key that as many shocks as observables.
  - Other issues:
    - → For heteroskedasticity: assumes that only volatility ratios changed, not shock propagation.
    - → Higher-order moments are hard to estimate, and analysis is necessarily fragile with respect to the estimated higher-order properties.

$\Longrightarrow$ To me reliance on explicitly defended economic identifying asn's is a virtue, not a bug.

Outline

1. The Time Series Identification Challenge

2. Invertibility + X
   - Exact Exclusion Restrictions
   - Sign Restrictions
   - Other Approaches
   - Degree of Invertibility

3. Identification without Invertibility
   - Instruments/Proxies
   - Recoverability

Aside: degree of invertibility

- Suppose we wish to use invertibility-based identification to study a **single shock** $\varepsilon_{j,t}$. This can work if and only if *that shock* is invertible:

$$\varepsilon_{j,t} \in \text{span}(y_\tau, -\infty < \tau \leq t) \Leftrightarrow \varepsilon_{j,t} = h' u_t \text{ for some } h$$

Weaker than invertibility of all shocks, referred to as partial invertibility. Forni-Gambetti-Sala (2019).

- But invertibility shouldn't be an either-or proposition. Intuitively, we should be able to do well if $\varepsilon_{1,t}$ is "**close to invertible**". How to formalize that?

$$\mathcal{R}^2_{j,0} \equiv 1 - \frac{\text{Var}^*(\varepsilon_{j,t} \mid \{y_\tau\}_{-\infty<\tau\leq t})}{\text{Var}(\varepsilon_{j,t})} = 1 - \text{Var}^*(\varepsilon_{j,t} \mid \{y_\tau\}_{-\infty<\tau\leq t})$$

  - Can show: the asymptotic bias of invertibility-based strategies is a function of $\mathcal{R}^2_{j,0}$, vanishing [if ID asn's are right] as $\mathcal{R}^2_{j,0} \to 1$ [e.g., see Plagborg-Møller & Wolf (2021)]
  - Appendix: what do we in general get in the non-invertible case? ▸ Details

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Identification without invertibility

- Recently there's been a push towards **invertibility-robust identification approaches**
- To me the most promising of these is the **narrative/macro-IV approach**
  - Assumes that, through institutional knowledge, the researcher can find a valid IV for a macro shock:
$$z_t = \alpha\varepsilon_{1,t} + \sigma_\nu \nu_t, \quad (\varepsilon_t', \nu_t)' \sim WN(0, I) \tag{6}$$
  - This brings macroeconometrics closer to microeconomic practice: try to find credible natural experiments that satisfy (6). We will thus focus on such IVs.
  - Main appeal: requires no further assumptions on the SVMA model, at least for IRFs
- Content of the next couple of slides:
  1. Discuss identification based on IVs (6) (including some popular examples)
  2. Briefly: sketch "recoverability"-based identification

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Setting & identification problem

- Let's suppose we have access to a **macro IV/proxy**:

$$z_t = \alpha\varepsilon_{1,t} + \sigma_\nu \nu_t, \quad (\varepsilon_t', \nu_t)' \sim WN(0, I)$$

  - How may we be able to find such z's?
    - → Institutional knowledge of policy decisions (e.g., fiscal spending $\perp$ to macro conditions)
      In fact you may more plausibly think that z_t correlates with many of the same *type* of shock, e.g., different monetary policy shocks. Remember Slide 28 in Lecture Note 4.
    - → High-frequency financial market responses (e.g., idiosyncratic global oil supply changes)
  - Even if found, such z's are unlikely to capture *all* of the hidden shock $\varepsilon_{1,t}$, so we'll invariably have measurement error ν_t

- **Q**: how can we use such IVs to identify our objects of interest?
  - Note that the econometric model is now the general SVMA + our IV equation:

$$\begin{aligned} y_t &= \Theta(L)\varepsilon_t, \\ z_t &= \alpha\varepsilon_{1,t} + \sigma_\nu \nu_t, \quad (\varepsilon_t', \nu_t)' \sim WN(0, I) \end{aligned}$$

Dynamic causal effects

- First observation: it's trivial to identify ***relative* dynamic causal effects = IRFs**
  - Note that we have

$$\text{Cov}(y_{i,t+\ell}, z_t) = \Theta_{i,1,\ell} \,\text{Cov}(\varepsilon_{1,t}, \alpha\varepsilon_{1,t} + \sigma_\nu \nu_t) = \alpha\Theta_{i,1,\ell}$$

  - Thus we can point-identify the IRF *ratio*

$$\frac{\Theta_{i,1,\ell}}{\Theta_{1,1,0}} = \frac{\text{Cov}(y_{i,t+\ell}, z_t)}{\text{Cov}(y_{1,t}, z_t)}$$

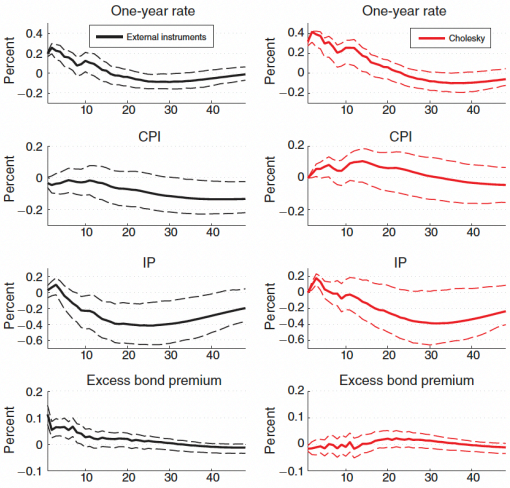
  - This is all we want: normalize nominal rate treatment path to increase by 100bp on impact
    [True even if z_t correlates with multiple types of shocks—get weighted av'g of treatment effects.]

- Note that none of these derivations assumed anything about **invertibility** (or more generally, our three sources of **indeterminacy**). Simply *not needed* for IV.

Examples of macro IVs

- Can we in practice hope to find such **macro IVs**? Two canonical examples:
  1. High-frequency **monetary policy surprises** [Gertler-Karadi (2015)]
     - Measure unexpected movements in interest rate futures around FOMC meeting dates
     - Identifying assumption: everyone knows rule $i_t = f(\Omega_t) + \varepsilon_t^m$ and information Ω_t, so surprise change must reflect ε_t^m [Potential problems? See literature on Fed information effect.]
     - Connection to Lecture Note 4: this averages across a lot of heterogeneous treatments
       Sometimes move short end of yield curve, sometimes long end.

High-frequency monetary shocks



First-stage regression:
F: 21.55; Robust F: 17.64; R^2: 7.76 percent; Adjusted R^2: 7.40 percent

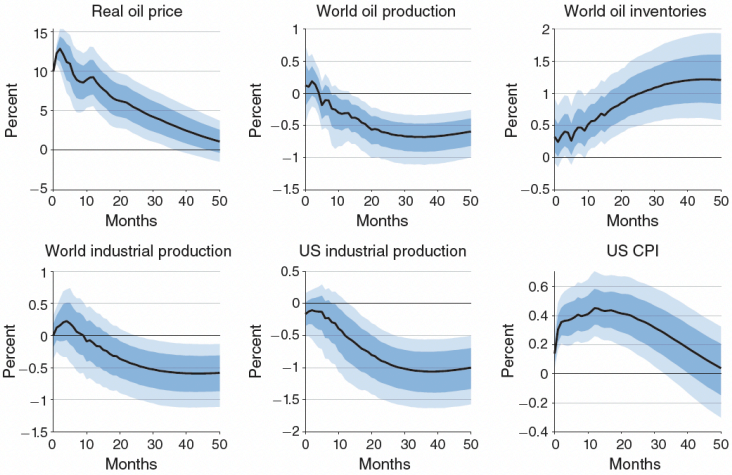
FIGURE 1. ONE-YEAR RATE SHOCK WITH EXCESS BOND PREMIUM

Examples of macro IVs

- Can we in practice hope to find such **macro IVs**? Two canonical examples:

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   - Measure unexpected movements in interest rate futures around FOMC meeting dates
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   - Connection to Lecture Note 4: this averages across a lot of heterogeneous treatments
     Sometimes move short end of yield curve, sometimes long end.
2. High-frequency **oil supply news** [Känzig (2021)]
   - Construction: measure unexpected movements in oil futures prices around OPEC meeting/announcement dates
   - Identifying assumption: surprise responses reflect *only* news about future oil supply
   - Estimate dynamic causal effects of *average* oil price increase = treatment path

High-frequency oil shocks



First-stage regression: F: 22.67, robust F: 10.55, R^2: 4.22%, Adjusted R^2: 4.04%

FIGURE 3. IMPULSE RESPONSES TO AN OIL SUPPLY NEWS SHOCK

SVAR-IV & invertibility

- Our IV analysis so far has proceeded without assuming **invertibility**. This is quite different from early time series work using **macro IVs**:
  - The popular "SVAR-IV" approach uses IVs as a way to solve the rotation problem in invertibility-based identification [This is actually what's done in Gertler-Karadi.]
  - Suppose invertibility holds, so $u_t = H\varepsilon_t$. We can then recover $\varepsilon_{1,t}$ as the projection of z_t on u_t (which is $\propto \varepsilon_{1,t}$), rescaled to have unit variance:

$$\mathbb{E}^*(z_t \mid u_t) = \mathbb{E}^*(z_t \mid \varepsilon_t) = \mathbb{E}^*(z_t \mid \varepsilon_{1,t}) = \alpha\varepsilon_{1,t}, \quad \alpha^2 = \text{Var}(\mathbb{E}^*(z_t \mid u_t))$$

  - Can then recover IRFs, the shock series, ... in the usual invertibility-based way

- But this all relied on **invertibility**, which we've seen is not needed

  It's not needed for IRFs. To get the shock, can get by with something intermediate—"recoverability". See the end of these slides, and Plagborg-Møller and Wolf (2022).

Testing invertibility

- With IVs we can in fact *test* invertibility. Key: under invertibility, we must have

$$\mathbb{E}^*(y_{i,t} \mid \{z_\tau, y_\tau\}_{-\infty<\tau<t}) = \mathbb{E}^*(y_{i,t} \mid \{y_\tau\}_{-\infty<\tau<t})$$

  - In words: z does not help to predict future y's above and beyond the info contained in y
  - Why? $z_t = \alpha\varepsilon_{1,t} + \sigma_\nu \nu_t$, where the second term is useless for forecasting future y, and the first term is captured by past y's. Suggests a simple test: does z Granger-cause y?

- If SVAR-IV is used even though **invertibility fails**, then we are subject to the issues reviewed in the slide appendix—bias that is a function of $\mathcal{R}^2_{1,0}$.

- Big picture: SVAR-IV is a bit of a "historical accident", reflecting path dependence
  - Literature thought in terms of identification through invertibility ("SVAR identification"), so just viewing instruments as a way of finding the right Q seemed natural

Aside: shock importance

- What can we say about **shock importance**, e.g. variance and historical decompositions?
  Issue: for this clearly need absolute impulse responses, and not just relative ones.
- Intuition on the **identification challenge**
  - IV consists of signal ($\alpha\varepsilon_{1,t}$) and noise ($\sigma_\nu\nu_t$). We don't know the signal-to-noise ratio.
  - This matters: e.g. can't know whether $\text{Corr}(y_{it}, z_t)$ is low because $\varepsilon_{1,t}$ is unimportant or because there's just a lot of noise
  - Formally, our challenge is to say something about the forecast variance ratio:

$$\begin{aligned}\text{FVR}_{i,1,h} &\equiv 1 - \frac{\text{Var}(y_{i,t+h} \mid \{y_{t-\ell}\}_{\ell=0}^{\infty}, \{\varepsilon_{1,t+\ell}\}_{\ell=1}^{\infty})}{\text{Var}(y_{i,t+h} \mid \{y_{t-\ell}\}_{\ell=0}^{\infty})} \\ &= \frac{\sum_{m=0}^{h-1} \frac{1}{\alpha^2} \text{Cov}(y_{i,t+h}, z_{t+h-m})^2}{\text{Var}(y_{i,t+h} \mid \{y_{t-\ell}\}_{\ell=0}^{\infty})}\end{aligned}$$

- Plagborg-Møller and Wolf (2022) prove that α and so FVR is **interval-identified** ▸ Details

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Recoverability

- Alternative approach: **recoverability-based identification**

Definition

A VMA process $\{y_t\}$ is said to be **recoverable** with respect to $\{\varepsilon_t\}$ if

$$\varepsilon_t \in \text{span}(y_\tau, -\infty < \tau < \infty)$$

- Thus the identification problem can be solved by looking into the past & future:

$$\varepsilon_t = \sum_{\ell=-\infty}^{\infty} \Phi_\ell y_{t-\ell} = \sum_{\ell=0}^{\infty} Q_\ell u_{t+\ell} = Q_0 u_t + Q_1 u_{t+1} + \dots$$

  - $\Phi(L)$ is a *two-sided* lag polynomial and $I = \sum_{\ell=0}^{\infty} Q_\ell \Sigma Q_\ell'$, $0 = \sum_{\ell=0}^{\infty} Q_\ell \Sigma Q_{h+\ell}' \ \forall h > 0$
  - ID is even harder: we are willing to call some n_y-dim. VMA "structural" (= assume away size indeterminacy), but still need to contend with static & *now dynamic* indeterminacy
    E.g.: Lippi-Reichlin (1994), Mertens-Ravn (2010), Chahrour-Jurado (2021)

Summary

- **This lecture note**: time series methods for identifying the SVMA Θ's
  - We began by characterizing the identification problem: indeterminacy from three sources
  - We then reviewed the solutions: invertibility + X, macro IVs, recoverability + X
- **Next**: discuss various popular econometric strategies to in practice implement these identification approaches & so estimate the Θ's

Appendix

Mis-identification without invertibility

- **Q**: what does invertibility-based identification recover without invertibility?
- We will derive everything from a **state-space representation**:

$$s_t = As_{t-1} + B\varepsilon_t \tag{7}$$
$$y_t = Cs_{t-1} + D\varepsilon_t \tag{8}$$

  - Invertibility is about the informativeness of y_t about s_t (and so ε_t)
  - We will compute this using the Kalman filter. Will rely on linear projections. Notation:

$$\hat{s}_{t|t} \equiv \mathbb{E}^* \left[s_t \mid \{y_\tau\}_{-\infty<\tau\leq t} \right]$$
$$\Sigma^s_{t|t} \equiv \text{var}^* \left[s_t \mid \{y_\tau\}_{-\infty<\tau\leq t} \right]$$

    and similarly for y_t

  - Algorithm: use standard linear projection formulas to update $\hat{s}_{t-1|t-1}$ and $\Sigma^s_{t-1|t-1}$ to $\hat{s}_{t|t}$ and $\Sigma^s_{t|t}$. Fixed point will give population limits (i.e., $t \to \infty$)

Kalman filtering algorithm

- Start with $\hat{s}_{t-1|t-1}$ and $\Sigma^s_{t-1|t-1}$. Predicting one period ahead:

$$\begin{aligned}
\hat{s}_{t|t-1} &= A\hat{s}_{t-1|t-1} \\
\Sigma^s_{t|t-1} &= A\Sigma^s_{t-1|t-1}A' + BB' \\
\hat{y}_{t|t-1} &= C\hat{s}_{t-1|t-1} \\
\Sigma^y_{t|t-1} &= C\Sigma^s_{t-1|t-1}C' + DD'
\end{aligned}$$

- Next we use y_t:

$$\begin{aligned}
\hat{s}_{t|t} = \hat{s}_{t|t-1} &+ \mathbb{E}\left[(s_t - \hat{s}_{t|t-1})(y_t - \hat{y}_{t|t-1})'\right] \times \\
&\mathbb{E}\left[(y_t - \hat{y}_{t|t-1})(y_t - \hat{y}_{t|t-1})'\right]^{-1}(y_t - \hat{y}_{t|t-1})
\end{aligned}$$

Kalman filtering algorithm

- Plugging in we get

$$\hat{s}_{t|t} = \hat{s}_{t|t-1} + \underbrace{\left[A\Sigma^s_{t-1|t-1}C' + BD'\right]\left[C\Sigma^s_{t-1|t-1}C' + DD'\right]^{-1}}_{\text{Kalman gain } K_t}(y_t - \hat{y}_{t|t-1})$$

  Similarly

$$\Sigma^s_{t|t} = \Sigma^s_{t|t-1} - \left[A\Sigma^s_{t-1|t-1}C' + BD'\right]\left[C\Sigma^s_{t-1|t-1}C' + DD'\right]^{-1}\left[A\Sigma^s_{t-1|t-1}C' + BD'\right]'$$

- Thus, given any history of the observables $\{y_t\}_{t=0}^T$, we can construct a sequence of estimates of the hidden states s_t

Innovations representation

- Letting $t \to \infty$, we obtain the steady-state Kalman gain and state uncertainty:

$$\begin{aligned} \Sigma^s &= (A - KC)\Sigma^s(A - KC)' + BB' + KDD'K' - BD'K' - KDB' \\ K &= (A\Sigma^s C' + BD')(C\Sigma^s C' + DD')^{-1} \end{aligned}$$

- This allows us to re-write the state-space system in **innovations form**:

$$\begin{aligned} \hat{s}_t &= A\hat{s}_{t-1} + K \underbrace{(y_t - \hat{y}_{t|t-1})}_{u_t} \\ y_t &= C\hat{s}_{t-1} + u_t \end{aligned}$$

  with $\Sigma^u = C\Sigma^s C' + DD'$

- Now we finally get to the payoff: write **Wold innovations** in terms of the ε_t's ...

Shocks and innovations

- The innovations representation gives

$$\begin{pmatrix} s_t \\ \hat{s}_t \end{pmatrix} = \begin{pmatrix} A & 0 \\ KC & A - KC \end{pmatrix} \begin{pmatrix} s_{t-1} \\ \hat{s}_{t-1} \end{pmatrix} + \begin{pmatrix} B \\ KD \end{pmatrix} \varepsilon_t$$

$$u_t = \begin{pmatrix} C & -C \end{pmatrix} \begin{pmatrix} s_{t-1} \\ \hat{s}_{t-1} \end{pmatrix} + D\varepsilon_t$$

to arrive at

$$u_t = \left\{ D + \begin{pmatrix} C & -C \end{pmatrix} \left(I - \begin{pmatrix} A & 0 \\ KC & A - KC \end{pmatrix} L \right)^{-1} \begin{pmatrix} B \\ KD \end{pmatrix} L \right\} \varepsilon_t = \sum_{\ell=0}^{\infty} M_\ell \varepsilon_{t-\ell}$$

  - Under invertibility, $s_t = \hat{s}_t$, so $u_t = D\varepsilon_t$, exactly as we have seen
  - Without invertibility, the identified shocks $\tilde{\varepsilon}_t = H^{-1} u_t$ are a linear combination of current and past true shocks ε_t
  - Can show that the weight on $\varepsilon_{j,t}$ is bounded above by $\sqrt{\mathcal{R}^2_{j,0}}$, and that SVAR-IV attains this bound Wolf (2020)

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Lower bound

- Finding a **lower bound** on shock importance is trivial
  - We must clearly have $\alpha^2 \leq \sigma_z^2$ – the IV can't be better than perfect
  - This upper bound on α gives a lower bound on the FVR, corresponding to naive regression of $y_{i,t+h}$ on z_t, ignoring measurement error
- In practice, IVs are noisy, so this lower bound is often **close to 0**

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Upper bound

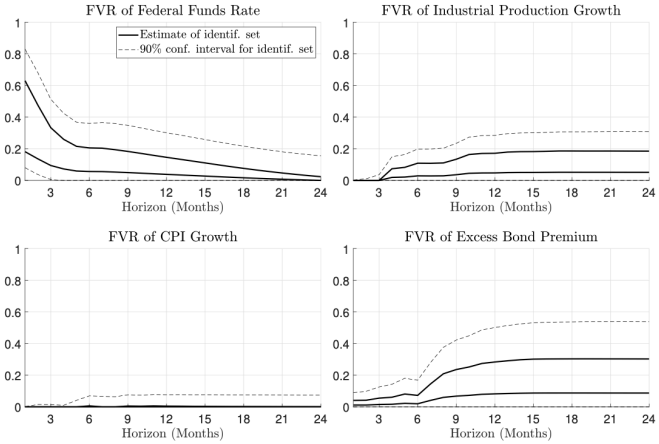
- Why should we expect there to be an **upper bound**? Example:
  - Suppose $y_{1,t}$ and $y_{2,t}$ have the same variance, but $\text{Cov}(y_{1,t}, z_t) \gg \text{Cov}(y_{2,t}, z_t)$. Then $\varepsilon_{1,t}$ can't explain much of $y_{2,t}$.
  - Why? If it explained much of $y_{2,t}$, it would *more than all of* $y_{1,t}$, which is impossible
- Let's formalize this intuition in a **static SVMA model**
  - The model now is $y_t = \Theta_0 \varepsilon_t$, and we let $\Sigma_y = \text{Var}(y_t) = \Theta_0 \Theta_0'$
  - Then we must have that

$$\Sigma_y - \frac{1}{\alpha^2} \text{Cov}(y_t, z_t) \, \text{Cov}(y_t, z_t)' \text{ is positive semi-definite} \tag{9}$$

  Why? otherwise the leftover part is not a valid stochastic process (e.g., negative variance)
- The full argument simply applies the requirement (9) to all **frequencies** separately

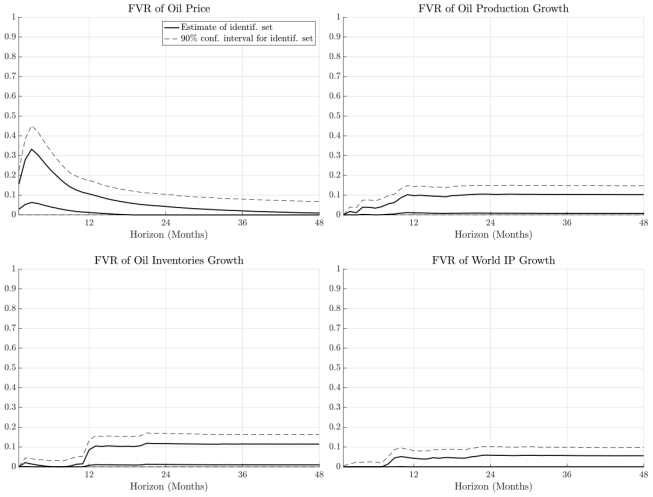
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High-frequency monetary shocks



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High-frequency oil shocks



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